

NAME

CUTEST_csj – CUTEst tool to evaluate constraint Jacobian values in a sparse format.

SYNOPSIS

CALL CUTEST_csj(status, n, X, nnzj, lj, J_val, J_var, J_con)

For real rather than double precision arguments, instead

CALL CUTEST_csj_s(...)

and for quadruple precision arguments, when available,

CALL CUTEST_csj_q(...)

DESCRIPTION

The CUTEST_csj subroutine evaluates the values of the Jacobian $J(x)$ of the constraint functions $c(x)$ of the problem decoded from a SIF file by the script *sifdecoder* at the point X in the constrained minimization case. The Jacobian is stored in a sparse format.

The problem under consideration is to minimize or maximize an objective function $f(x)$ over all $x \in R^n$ subject to general equations $c_i(x) = 0$, ($i \in 1, \dots, m_E$), general inequalities $c_i^l \leq c_i(x) \leq c_i^u$ ($i \in m_E + 1, \dots, m$), and simple bounds $x^l \leq x \leq x^u$. The objective function is group-partially separable and all constraint functions are partially separable.

ARGUMENTS

The arguments of CUTEST_csj are as follows

status [out] - integer

the output status: 0 for a successful call, 1 for an array allocation/deallocation error, 2 for an array bound error, 3 for an evaluation error,

n [in] - integer

the number of variables for the problem,

X [in] - real/double precision

an array which gives the current estimate of the solution of the problem,

nnzj [out] - integer

the number of nonzeros in J_val,

lj [in] - integer

the actual declared dimensions of J_val, J_var and J_con,

J_val [out] - real/double precision

an array which gives the values of the nonzeros of the general constraint functions evaluated at X . The i -th entry of J_val gives the value of the derivative with respect to variable J_var(i) of constraint function J_con(i),

J_var [out] - integer

an array whose i -th component is the index of the variable with respect to which J_val(i) is the derivative,

J_con [out] - integer

an array whose i-th component is the index of the constraint function of which J_val(i) is the derivative,

AUTHORS

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SEE ALSO

CUTEst: a Constrained and Unconstrained Testing Environment with safe threads,

N.I.M. Gould, D. Orban and Ph.L. Toint,

Computational Optimization and Applications **60**:3, pp.545-557, 2014.

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N.I.M. Gould, D. Orban and Ph.L. Toint,

ACM TOMS, **29**:4, pp.373-394, 2003.

CUTE: Constrained and Unconstrained Testing Environment,

I. Bongartz, A.R. Conn, N.I.M. Gould and Ph.L. Toint,

ACM TOMS, **21**:1, pp.123-160, 1995.

cutest_ccfsg(3M), sifdecoder(1)